Coursework Submission for CSN11127

Edinburgh Napier University 2025-2026 Trimester 2

TOPIC NO(1-7): 7

TITLE: Large Language Model Assisted Reconnaissance

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1. Introduction

Penetration testing is an authorised security assessment used to identify weaknesses before they are exploited by malicious actors (Constantin, 2023). This coursework focuses specifically on the reconnaissance phase of the test, where information such as live hosts, open ports, service banners, operating system and other information is collected. Reconnaissance is significant because it provides information for a later stage like vulnerability analysis , exploit development and others (Malatiji et al., 2024)

The aim of this coursework is to evaluate, as a preliminary proof of concept, whether Claude Desktop integrated with Kali Linux through Model Context Protocol (MCP) can support structured reconnaissance in a controlled virtual lab. To achieve the aim the experiment, requires use of Claude Desktop on a Windows VM client, Kali Linux as the reconnaissance machine server, and Metasploitable2 as the authorised target inside an isolated VMware host-only network.

The scope is strictly limited to reconnaissance and attack surface intelligence gathering using the Claude. The practical work includes host discovery, port scanning, service enumeration, operating system fingerprinting and web technology identification. It does not include exploitation, privilege escalation, persistence, post-exploitation, malware, password attacks or testing against external systems.

2. Literature Review and Methodology

2.1 Literature Review

Nila and Patriciu (2023) describe reconnaissance as a key early phase of penetration testing and identify tools such as Nmap, Gobuster, Burp Suite, OWASP ZAP, Shodan, Recon-ng and theHarvester as common reconnaissance tools. The manual regex method detected 416 suspicious events while the Large Language Model (LLM) based approach identified 3,699 suspicious entries indicating that LL M can detect broader contextual patterns. However, the risk of over calcification and false positive are the consequence of using LLM which still requires human validation (Nila and Patriciu, 2023).

Artificial intelligence(AI) assisted penetration testing is an emerging area of cybersecurity research however the literature shows that the field is still immature. Systematic mapping from 57 studies on AI-driven penetration testing found that reinforcement learning, deep learning, supervised learning and generative AI are the main techniques used for automating task such as service enumeration, vulnerability assessment, networking scanning and reporting. AI can improve automation and vulnerability identification, but it also identifies challenges in integration difficulty, scalability limitations, ethical concerns and misuse risks (Alwabisi, 2025)

The role of LLMs in penetration testing has been examined by comparing Claude Opus, GPT-4 Turbo and Copilot across the Penetration Testing Execution Standard (PTES) in a controlled virtualised environment. The finding was that Claude Opus retained context , adapted when commands failed and generated more context specific guidance than the alternative. However, the study also reinforces that human judgement remains necessary for the command accuracy and the safe execution. (Martínez et al., 2025).

The integration on the AI model into the machine requires high skill and experience which itself is a challenge, nevertheless Anthropic has introduced and open standard connector know as Model context protocol (MCP) which connects AI assistants to external systems with a standardised two-way protocol between MCP client and MCP server. For the experiment the MCP has been used to integrate windows layer between Claude desktop and kali Linux. The MCP supports a multi tools integration and orchestration instead of building a unique connector for every tool. However, the security risk of LLM assisted workflow constructs a threat taxonomy

across the pipeline as there is a need for creation, deletion and maintaining of the server (Hou et al., 2025)

The literature indicates that LLMs can support reconnaissance through command generation, tool coordination and output interpretation. Nevertheless, they remain unsuitable for unsupervised use because of hallucination, context loss and potentially unsafe recommendations. A key gap is that most studies focus on wider automation or exploitation, while reconnaissance is rarely evaluated as a standalone phase this coursework addresses that gap through a controlled LLM-assisted recon lab.

Related work such as Goya et al (2024), Temara (2023) and Luong et al. (2025) supports the idea that LLM can assist reconnaissance, tool orchestration and pentesting workflows.

2.2 Literature Selection Methodology

A PRISMA-inspired literature selection method was followed to make the literature review more systematic and repeatable. The search focused on LLMs, penetration testing, reconnaissance, AI-assisted tooling and Penetration testing automation.

The databases used were Google Scholar, IEEE Xplore, ACM Digital Library, arXiv, ScienceDirect and Elsevier. Search terms included "Large Language Model", "LLM", "Generative AI", "penetration testing", "AI-assisted penetration testing", "reconnaissance", "web reconnaissance", "attack surface discovery", "Kali Linux", "Claude", "Model Context Protocol" and "PentestGPT".

The main search strings were "LLM" OR "Large Language Model" AND "reconnaissance" , "penetration testing" OR "ethical hacking", "Generative AI" OR "Claude" OR "GPT" AND "penetration testing". As seen in below table1 inclusion criteria for the papers were language in English, published mainly within the last four years, and directly relevant to AI, LLMs, reconnaissance, penetration testing or tool integration for penetration testing. Exclusion criteria were duplicate records, blog, unrelated to cybersecurity, focused purely on malware, other LLM models.

PRISMA Stage	**No of articles**
Records identified	86
Duplicates removed	14
Records screened	72
Records excluded after title/abstract	45
Full-text papers assessed	27
Full-text papers excluded	15
Final papers included	10

Table 1: Paper selection stages (author, 2026)

The selected papers were then analysed thematically. Five themes were identified: AI in penetration testing, LLM-assisted reconnaissance, LLM architecture and tool integration, LLM selection for penetration testing.

3. Experimental Activities and Results

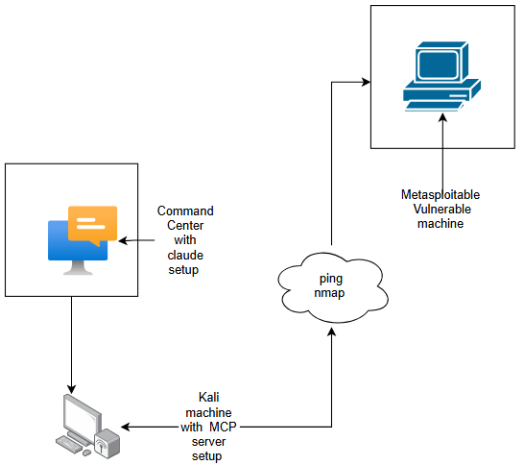


Figure 1: LLM assisted lab architecture (author, 2026)

As seen on figure 1 and briefed on table 2 the experiment was carried out using three virtual machines in an isolated VMware host. To set up the lab the Claude desktop, Kali OS and Metasplotable2 was downloaded from their respective official site. For the integration of the Claude with the windows and the windows machine with the kali OS MCP was used.

Machines	**Role**	**IP Address**	**Details**
Claude Desktop Client	LLM assisted command interface	Windows VM	Issued all reconnaissance instructions via natural language prompts to MCP
Kali Linux 2024 Server	Reconnaissance machine	192.168.5.130	Executed tools through MCP
Metasploitable2	Target	192.168.5.132	Deliberately vulnerable Linux tested across all 3 DVWA security levels
VMware Workstation	Hypervisor	Host Machine	Host-only networking .Fully isolated,

Table 2: The machines used in the lab and its details (author, 2026)

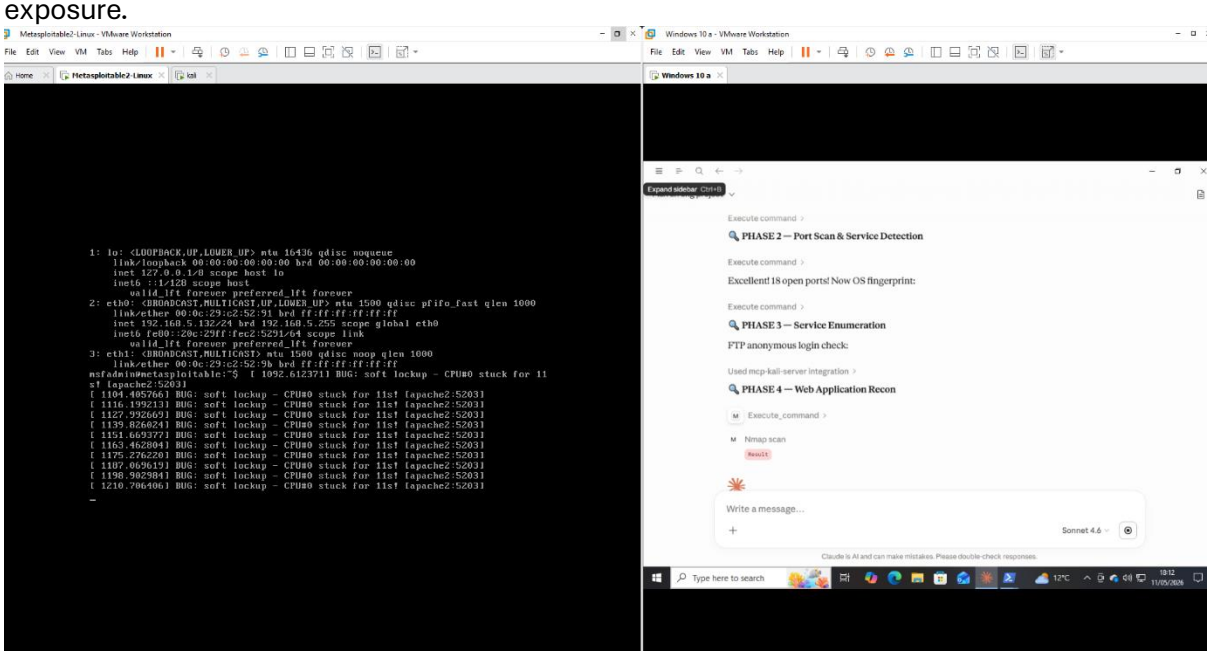
The pipeline has three important properties. Firstly, the natural language was used where no command construction is required. Secondly the Claude went into adaptive mode and modified scan parameters in real time by reducing thread counts to avoid crashing the VM and lastly the raw output was contextualised flagging CVEs and assessing the severity without the additional prompting. The experiment began with a simple prompt Recon the target 192.168.5.132.

Recon Area	**Tools / Technique**	**Key Findings**	**Low**	**Mid**	**High**
Host and OS discovery	nmap -sn, ping, nmap -O	Target live; hostname and Linux OS fingerprint disclosed	Yes	Yes	Yes
Port and service enumeration	nmap -sV -T4	23 open ports; all services disclosed full banners	23	23	23
FTP enumeration	ftp-anon, ftp-syst	vsftpd 2.3.4; anonymous access; CVE-2011-2523 indicator	Yes	Yes	Yes
SMB enumeration	smb-enum-shares, smb-enum-users	Samba 3.0.20; anonymous read/write; 35 users identified	35	35	35

User enumeration	Finger, SMTP VRFY/EXPN	User details leaked and accounts corroborated across services	Yes	Yes	Yes
SNMP and NFS enumeration	snmpwalk, onesixtyone, showmount	Default SNMP community; world-readable NFS exports	Yes	Yes	Yes
Database reconnaissance	MySQL/PostgreSQL NSE scripts	MySQL empty root password; PostgreSQL version disclosed	Yes	Yes	Yes
Web application mapping	gobuster, whatweb, nikto, curl	phpinfo, phpMyAdmin, WebDAV, Tomcat Manager, HTTP TRACE exposed	Yes	Yes	Yes
Web hardening checks	Header inspection / Nikto	0/5 key HTTP security headers present	0/5	0/5	0/5
Critical vulnerability indicators	Service/version correlation	15 critical CVE or misconfiguration indicators identified	15	15	15

Table 3: Results of reconnaissance from low, mid and high security level (author, 2026)

As shown on the above table 3 after the experiment, the test identified a consistent attack surface across all three Damn Vulnerable Web App (DVWA) security levels. In each configuration, Metasploitable2 exposed 23 open ports, full-service banners, operating system information, user enumeration channels, weak file-sharing services, exposed database services and multiple web application weaknesses. Key indicators included vsftpd 2.3.4, Samba 3.0.20, Telnet, SNMP with a default community string, world-readable NFS exports, MySQL with an empty root password, phpinfo exposure, phpMyAdmin access, WebDAV and Tomcat Manager exposure.



Picture 1: DVWA crashed during the web app recon (author, 2026)

During the web application reconnaissance, the Metasploitable 2 showed a Linux soft lockup message displayed on the target console. This indicated that the target was crashed as result of

aggressive scans. The prompt was then refined to " Recon the target 192.168.5.130" without crashing, after which the command was adapted, and the scans were run on low intensity.

The main result is that DVWA Low, Medium and High settings did not change the network level reconnaissance findings. This indicates that DVWA security levels affect application layer behaviour inside DVWA, but do not reduce the underlying host attack surface. Also the crashed out come suggest that the human supervision and scan intensity control is required during the prompt generation and test process.

4. Evaluation and Discussion

The experiment demonstrates that Claude Desktop integrated with Kali Linux through MCP can support reconnaissance by structuring tasks, selecting appropriate tools and interpreting outputs. This supports Martinez et al., (2025) finding that generative AI tools are useful for specific penetration testing tasks, although they cannot fully automate the entire process.

The main strength of the LLM-assisted workflow was organisation. Claude helped move from host discovery to service enumeration and then to web reconnaissance in a logical sequence. This is consistent with PentestGPT's finding that LLMs can use tools, interpret outputs and propose next actions (Deng et al., 2024). In practical terms, this reduces the burden of command creation and allows tester to focus on the findings.

However, the results also show that LLM-assisted reconnaissance must be supervised. Metasploitable2 is intentionally vulnerable, and aggressive scans can destabilise the VM hence the prompt must be filtered through human judgement. Den et al., (2026) argue that LLM agents can fail because of context overload, poor planning and inability to estimate task difficulty. The experiment was limited by the scope which helped to limit the risk of crashing.

The experiment highlighted both opportunity and risk. The opportunity is that LLMs can make reconnaissance workflows more accessible, repeatable and well documented as achieved from the experiment conducted by the Claude, MCP and kali workflow. Whereas the risk is that this lowers the technical barrier for misuse. Therefore, any LLM assisted penetration testing workflow should include scope control, authorisation requiring human approval before command execution, and strict separation between reconnaissance and exploitation unless explicitly authorised.

5. Conclusion

This coursework critically evaluated the use of Claude Desktop integrated with Kali Linux through MCP for reconnaissance focused penetration testing. The experiment showed that the LLM assisted workflow could support host discovery, service enumeration, web reconnaissance and attack surface intelligence gathering in a controlled virtual lab.

The practical findings showed that Metasploitable2 exposed 23 open ports and multiple high-risk services across all DVWA security levels. The most important technical finding was that DVWA Low, Medium and High settings did not change the network-level attack surface. This demonstrates that DVWA security levels affect web application logic, not the underlying operating system or network service exposure.

Finally, LLMs are valuable for reconnaissance planning, command explanation and result interpretation. However, they require human oversight because of risks including hallucination, unsafe recommendations, context loss and scan intensity errors.

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